BEE 4750 Homework 2: Systems Modeling and Simulation

Name:

ID:

> **Due Date**
>
> Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```julia
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/BEE 4750 /HW 2/hw2-regina`
   Installed Fontconfig_jll    v2.17.1+0
   Installed Libmount_jll     v2.41.1+0
   Installed WorkerUtilities    v1.6.1
   Installed JpegTurbo_jll     v3.1.3+0
   Installed Accessors       v0.1.42
   Installed Preferences      v1.5.0
   Installed Roots         v2.2.10
   Installed DataFrames      v1.8.0
   Installed WeakRefStrings    v1.4.2
   Installed DataStructures    v0.19.1
   Installed ConstructionBase   v1.6.0
   Installed Expat_jll       v2.7.1+0
   Installed Ghostscript_jll    v9.55.1+0
   Installed StatsBase       v0.34.6
   Installed CompositionsBase   v0.1.2
   Installed CommonSolve      v0.2.4
   Installed InverseFunctions   v0.1.17
   Installed Plots         v1.40.19
   Installed PrettyTables     v3.0.8
   Installed Libuuid_jll      v2.41.1+0
   Installed OpenSSL_jll      v3.5.2+0
   Installed Latexify       v0.16.10
   Installed Glib_jll       v2.86.0+0
   Installed FilePathsBase     v0.9.24
   Installed SortingAlgorithms   v1.2.2
   Installed CSV          v0.10.15
Precompiling project...
    538.4 ms   WorkerUtilities
    480.1 ms   CompositionsBase
    482.0 ms   InverseFunctions
    479.6 ms   ConstructionBase
    481.7 ms   CommonSolve
    596.5 ms   Preferences
    693.6 ms   FilePathsBase
    687.2 ms   LogExpFunctions → LogExpFunctionsInverseFunctionsExt
    693.9 ms   InverseFunctions → InverseFunctionsDatesExt
    733.6 ms   Unitful → InverseFunctionsUnitfulExt
    811.3 ms   InverseFunctions → InverseFunctionsTestExt
   1313.9 ms   DataStructures
    870.2 ms   CompositionsBase → CompositionsBaseInverseFunctionsExt
    495.7 ms   ConstructionBase → ConstructionBaseLinearAlgebraExt
    436.6 ms   Unitful → ConstructionBaseUnitfulExt
```

```
 623.0 ms   PrecompileTools
 700.0 ms   JLLWrappers
 940.5 ms   FilePathsBase → FilePathsBaseMmapExt
 986.4 ms   SortingAlgorithms
1417.3 ms   FilePathsBase → FilePathsBaseTestExt
 650.5 ms   Graphite2_jll
 973.2 ms   OpenSSL_jll
1831.8 ms   RecipesBase
1882.8 ms   StringManipulation
 485.8 ms   Libmount_jll
 466.9 ms   EpollShim_jll
 424.7 ms   LLVMOpenMP_jll
 448.6 ms   Bzip2_jll
 401.3 ms   Xorg_libICE_jll
 399.4 ms   Xorg_libXau_jll
2746.1 ms   Accessors
 464.2 ms   libpng_jll
 405.2 ms   libfdk_aac_jll
 401.4 ms   LAME_jll
 391.6 ms   LERC_jll
 426.3 ms   fzf_jll
 368.2 ms   mtdev_jll
 383.4 ms   Ogg_jll
 441.7 ms   JpegTurbo_jll
 475.9 ms   XZ_jll
 389.6 ms   Xorg_libXdmcp_jll
 422.8 ms   x265_jll
 423.2 ms   x264_jll
 447.9 ms   libaom_jll
 450.7 ms   Zstd_jll
 411.8 ms   Expat_jll
3469.1 ms   ColorSchemes
 459.0 ms   Opus_jll
 472.6 ms   LZO_jll
 462.9 ms   Xorg_xtrans_jll
 445.8 ms   libevdev_jll
 532.0 ms   Libiconv_jll
 494.1 ms   Libffi_jll
 425.2 ms   Libuuid_jll
 438.7 ms   eudev_jll
 469.5 ms   FriBidi_jll
 344.9 ms   Xorg_libSM_jll
 875.1 ms   Pixman_jll
```

```
1060.9 ms   FreeType2_jll
1136.9 ms   Accessors → LinearAlgebraExt
1984.1 ms   StatsBase
 773.6 ms   Accessors → TestExt
 648.0 ms   Accessors → UnitfulExt
 479.7 ms   Xorg_libxcb_jll
 882.2 ms   libvorbis_jll
 959.4 ms   JLFzf
3154.8 ms   OpenSSL
 448.6 ms   Dbus_jll
1768.2 ms   Ghostscript_jll
 475.9 ms   Wayland_jll
 608.9 ms   libinput_jll
1192.6 ms   GettextRuntime_jll
2053.0 ms   Libtiff_jll
8805.7 ms   Parsers
 360.1 ms   Xorg_xcb_util_jll
 335.2 ms   Xorg_libX11_jll
2223.8 ms   Fontconfig_jll
2630.7 ms   Roots
2079.4 ms   Glib_jll
1069.2 ms   InlineStrings → ParsersExt
 368.3 ms   Xorg_xcb_util_image_jll
 337.3 ms   Xorg_xcb_util_keysyms_jll
 378.4 ms   Xorg_xcb_util_renderutil_jll
 481.6 ms   Xorg_xcb_util_wm_jll
1131.6 ms   JSON
 681.6 ms   Xorg_libXrender_jll
3602.8 ms   Latexify
 692.7 ms   Xorg_libXext_jll
 636.0 ms   Xorg_libXfixes_jll
 556.0 ms   Xorg_libxkbfile_jll
 584.5 ms   Xorg_xcb_util_cursor_jll
 926.9 ms   Roots → RootsUnitfulExt
 907.5 ms   WeakRefStrings
 427.3 ms   Xorg_libXinerama_jll
 443.8 ms   Xorg_libXrandr_jll
 790.3 ms   Latexify → SparseArraysExt
 437.9 ms   Libglvnd_jll
7422.9 ms   PlotUtils
1317.6 ms   UnitfulLatexify
 428.5 ms   Xorg_libXcursor_jll
 409.0 ms   Xorg_libXi_jll
```

```
   400.6 ms    Xorg_xkbcomp_jll
  1166.8 ms    Xorg_xkeyboard_config_jll
   371.8 ms    xkbcommon_jll
   557.2 ms    Vulkan_Loader_jll
  2378.4 ms    PlotThemes
  3760.9 ms    RecipesPipeline
  4986.4 ms    Cairo_jll
  1403.2 ms    HarfBuzz_jll
  1220.8 ms    libass_jll
  1243.5 ms    Pango_jll
  4851.7 ms    Qt6Base_jll
   403.9 ms    libdecor_jll
   432.3 ms    GLFW_jll
 13871.3 ms    HTTP
  1057.0 ms    Qt6ShaderTools_jll
  2684.2 ms    FFMPEG_jll
  1574.2 ms    Qt6Declarative_jll
   358.1 ms    Qt6Wayland_jll
   747.7 ms    FFMPEG
  1232.9 ms    GR_jll
 21617.3 ms    PrettyTables
 11840.1 ms    CSV
  2247.9 ms    GR
 24016.3 ms    DataFrames
  1459.1 ms    Latexify → DataFramesExt
 33918.9 ms    Plots
  2830.6 ms    Plots → UnitfulExt
 128 dependencies successfully precompiled in 67 seconds. 91 already precompiled.
```

```julia
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO2 concentrations, and ocean CO2 concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

```
"""
                    (+)
Atmospheric CO         Global Mean Temperature

                                 (-)
                                |
                           Ocean CO

       (-)         -
"""
```

```
"                    (+)\nAtmospheric CO        Global Mean Temperature\n
```

Problem 1 SOLUTION:

Henry's Law says that solubility of CO in seawater decreases as temperature increases. In other words, warm water holds less CO . Because of this, the arrow pointing from global mean temperature to the concentration of CO in the ocean is a negative (-) sign and this is a dampening relationship. As temperature increases, the ocean's water warms and CO concentrations decrease.

When concentrations of CO in the ocean increase, the concentration of CO in the atmosphere decrease since the ocean is acting as a carbon sink and taking out CO from the atmosphere. Because of this, the sign is negative (-) and this is a dampening relationship.

Finally, when CO concentrations in the atmosphere increases, the greenhouse effect is magnified since the CO is now absorbing heat and re-emitting it. As a result, the global mean temperature to increase. This is an amplifying relationship and the arrow's sign is positive (+).

This suggests that the overall feedback between temperature and the ocean carbon cycle is an amplyfing (or positive) feedback because an increase in atmospheric CO , increases temperature, which creates less CO2 in the ocean, which means less CO is being absorbed by the ocean and there is more of it in the atmosphere, which finally leads to even more warming.

> **Tip**
>
> Think about Henry's law for CO2.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/outflows between these boxes. How did you decide how many boxes were needed?

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

> **Tip**
>
> Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system in compliance with a regulatory limit of 2.3 kg/(1000m^3). You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized "Snowball Earth". In this problem, we'll introduce a simple model of the Earth's energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C\frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}} + \underbrace{a \ln\left(\frac{[CO_2]}{[CO_2]_{PI}}\right)}_{\text{greenhouse effect}}, \tag{1}$$

where:

- T is the Earth's global mean temperature (in $^\circ$C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be 51 J/m^2/$^\circ$C;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproteorozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3$ W/m^2/$^\circ$C (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2$ W/m^2 (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1$ yr.

Problem 3.2

Rather than assuming a constant value for the albedo α, we will represent the ice-albedo feedback by letting α depend on T:

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T+10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoprotereozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C?

References

List any external references consulted, including classmates.